

Guesstimate Case Study

Estimating the Annual Sale of Domestic Refrigerators in India

1. Problem Statement

Estimate the number of refrigerators sold in India every year.

2. Clarifying Question

Interviewee: Should we consider domestic refrigerators only, or include commercial refrigerators too?

Interviewer: For this question, consider only domestic refrigerators.

3. Approach

Approach used: Demand-Side (Top-Down) Approach

Guesstimates like this can typically be solved using a demand-side approach (estimating end-consumer need and working down to units), a supply-side approach (estimating from manufacturer/dealer production and sales data), or a bottom-up approach (building up from a small, known unit and scaling). This solution uses the demand-side approach — first estimating the total addressable market (i.e., total refrigerators currently owned across Indian households) by working top-down from population, and then converting this stock figure into annual unit sales by accounting for replacement demand and new/growth demand.

- Why demand-side: household and income-class data (population, family size, income distribution) is more reliable to assume than manufacturer shipment data, making it the natural default for consumer durables like refrigerators.

4. Step 1: Total Households in India

Parameter	Value
Total population of India	130 crore
Average family size	4 members
Total households	$130 / 4 = 32.5$ crore

5. Step 2: Segmenting Households

Households are split by geography (Metro/Tier I, Tier II/III, and Villages), and within each geography, by income class (Poor, Middle-class, Rich).

Segment	Poor %	Middle-class %	Rich %
Metro / Tier I	20%	60%	20%
Tier II / III	40%	50%	10%
Villages	60%	38%	2%

Assumption on household split across geographies: Metro/Tier I = 20% of households, Tier II/III = 15%, and Villages = 65%.

6. Step 3: Refrigerator Penetration Rate

Penetration reflects the assumption that poor households do not own refrigerators, while ownership varies for middle-class and rich households by geography.

Segment	Poor	Middle-class	Rich
Metro / Tier I	0%	50%	100%
Tier II / III	0%	40%	100%
Villages	0%	0%	100%

7. Step 4: Market Size Calculation

Segment	Households (crore)	Effective Penetration	Refrigerators (crore)
Metro / Tier I	$0.20 \times 32.5 = 6.50$	50%	3.25
Tier II / III	$0.15 \times 32.5 \approx 5.00$	30%	1.50
Villages	$0.65 \times 32.5 \approx 21.00$	2%	0.42

Total Market Size (Refrigerator Stock) = $3.25 + 1.50 + 0.42 \approx 5.17$ crore refrigerators

8. Step 5: Converting Market Size to Annual Sales

Annual unit sales are not the same as total market size. Two demand components drive yearly sales:

- Replacement demand — households replacing an existing refrigerator that has reached the end of its useful life.
- New / growth demand — first-time buyers, driven by rising incomes and household formation.

Assumption	Value
Average life of a refrigerator	10 years
Annual market growth rate	5%

Formula

Refrigerators sold per year = (Market size $\div$ Average life) + (Growth rate $\times$ Market size)

9. Final Calculation

Component	Calculation	Value (crore)
Replacement demand	$5.17 / 10$	0.517
New / growth demand	5.17×0.05	0.259
Total annual sales	$0.517 + 0.259$	≈ 0.775

Estimated refrigerators sold in India every year ≈ 0.775 crore ≈ 77.5 lakh units

10. Key Takeaways for Presentation

- Always begin by clarifying scope (domestic vs. commercial) before diving into numbers.
- Explicitly name the approach (demand-side / supply-side / bottom-up) before starting the calculation — it signals structured thinking to the interviewer.
- Structure the estimate top-down: Population → Households → Segments → Penetration → Market size.
- Clearly separate 'market size' (stock) from 'annual sales' (flow) — this distinction is often what interviewers probe on.
- State every assumption explicitly (family size, growth rate, product life) so the interviewer can challenge or adjust them.
- Round numbers at each step to keep the mental math fast and the story easy to follow.